

# Session Layer

- Introduction
- Basic Responsibilities
- Protocols

# Session Layer

- the session layer provides the mechanism for opening, closing and managing a session between end-user application processes
- provides the mechanism for controlling the dialogue between applications in end systems.
- In many cases, there will be little or no need for session-layer services, but for some applications, such services are used.

# key services

## 1. Dialogue Discipline

- This can be two-way simultaneous (full duplex) or two way alternate (half duplex) communication between processes.
- In communication Each host plays dual roles they are requesting service, like a client and replaying with service ,like a server. determining which role they are playing at any given moment is called dialogue control.

# Contd.

## 2. Token Management:

- it prevents the two users to simultaneously attempt the same critical operation.

# Contd.

## 3. Recovery

- The session layer can provide a check pointing (synchronization point) mechanism, so that if a failure of some sort occurs between checkpoints, the session entity can retransmit all data since the last checkpoint.

# Popular Session Layer Protocols

The primary job of session layer protocols is to provide the means necessary to **set up, manage, and end sessions**

- **ADSP:** AppleTalk Data Stream Protocol
- **PAP :** Password Authentication Protocol
- **PPTP:** Point-to-Point Tunneling Protocol
- **RPC:** Remote Procedure Call Protocol
- **RTCP:** Real-time Transport Control Protocol

# Presentation Layer

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# Presentation Layer

- The presentation layer is concerned with the **syntax and semantics** of the information exchanged between two systems.

# Specific Responsibilities

## 1. Translation

- different computers use different encoding systems
- The presentation layer at the sender changes the information from its **sender-dependent format into a common format**. At the receiving machine changes the common format into its receiver-dependent format.

# Contd.

## 2. Encryption

- To carry sensitive information, a system must be able to ensure privacy.
- the sender transforms the original information to another form and sends the resulting message out over the network.
- **Decryption** reverses the original process to transform the message back to its original form.

# Contd.

## 3. Compression

- Data compression reduces the number of bits contained in the information.
- Data compression becomes particularly important in the transmission of multimedia such as text, audio, and video.

# Application Layer

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# Application Layer

- Enables the user, whether human or software, to **access the network**.
- It provides **user interfaces** and **support for services** such as **electronic mail, remote file access and transfer, shared database management**, and other types of distributed information services.

# Contd.

- Typical application layer protocols
  - Domain Name System (DNS)
  - Hyper Text Transfer Protocol (HTTP)
  - File Transfer Protocol (FTP)
  - E-mail (SMTP,POP,IMAP)
  - DHCP Services

# 1. Domain Name System (DNS)

- It would be impossible to remember all of the IP addresses
- DNS provides a way for hosts to use this name to request the IP address of a specific server.
- DNS names are registered and organized on the Internet within specific high level groups, or domains.
  - Some of the most common high level domains on the Internet are .com, .edu, and .net
- A DNS server contains a table that associates hostnames in a domain with corresponding IP addresses

## 2. Web client and web server

- A web client first receives the IP address of a web server from DNS server Then the client browser uses that IP address and port 80 to request web services
- This request is sent to the server using the Hypertext Transfer Protocol (HTTP)
- The information content of a web page is encoded using specialized 'mark-up' languages.
  - E.g. HTML (Hypertext Mark-up Language)

# 3. File Transfer Protocol (FTP)

- Allows users to transfer files
- file management functions (file uploads ,downloads)
- FTP service uses two different ports to communicate between client and server
  - Requests to begin an FTP session are sent to the server using destination port 21.
  - Once the session is opened, the server will change to port 20 to transfer the data files
- FTP client software is built into computer operating systems and into most web browsers

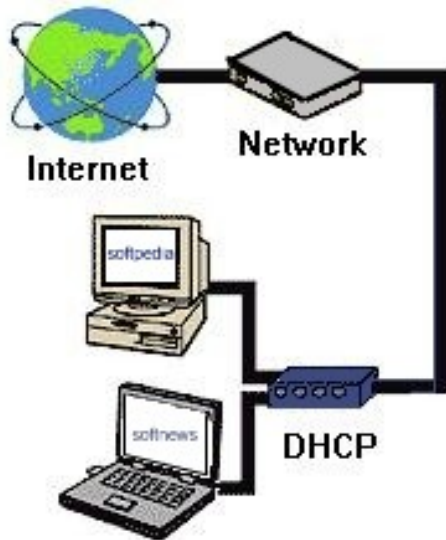
## 4. Email Servers (SMTP, POP3 and IMAP4)

- Each mail server receives and stores mail for users who have mailboxes configured on the mail server
- Each user with a mailbox must then use an email client to access the mail server and read these messages
  - Mailboxes are identified by the format:  
**user@company.domain**

# Contd.

- Three application protocols used in processing email include
  1. **Simple Mail Transfer Protocol (SMTP)**
    - to send mail from client to server or server to server
  2. **Post Office Protocol (POP3)**
    - to download email from server to client, and the server deletes the mail
  3. **Internet Message Access Protocol (IMAP4)**
    - to download email from server to client, and the server does not delete (keeps) the mail

# 5. Dynamic Host Configuration Protocol (DHCP)



- It is a method for assigning IP addresses permanently or to individual computers in an organization's network
- DHCP lets a network administrator supervise and distribute IP addresses from a central point and automatically sends a new IP address when a computer is plugged into a different place in the network .

**Thank You**